Activity No. 1 Getting Started with React Native	
Course Code: CPE 025B	**Program:** Computer Engineering
Course Title: Software Design 2	**Date Performed: 03/27/26**
Section: CPE32S1	**Date Submitted: 03/27/26**
Name: Edson Ray San Juan	**Instructor: Engr. Neil Matira**

1. Objective(s)

This activity aims to enable students to build a basic application with local state management and Create React Native App.

2. Intended Learning Outcomes (ILOs)

After this module, the student should be able to:

- Configure a React Native development using the Expo Go environment;
- Deploy a live-demo of a mobile application on both iOS and Android.

3. Discussion

There are two common ways to set up a React Native development environment: expo-cli and react-native CLI.

expo-cli

This is a command-line utility for configuring and publishing React Native projects. This is the best way to get started as a beginner. This tool generates a QR code which you can scan to launch the app on your device. As you update your code, the changes will automatically be reflected on your device. In order to preview the app on your device, you'll be prompted to download the Expo app, which is a React Native app previewing client.

The downside to this approach is that it only works for pure-JavaScript apps. If at any point you need to use native modules (Swift, Kotlin, etc), you'll need to eject. The expo-cli utility provides an eject command to export your app into the same format used by react-native CLI, which you can then add native modules to.

The company behind expo-cli, Expo, has been heavily involved in the React Native codebase and community since it's inception, so you can be confident that the tool is well-maintained and doesn't deviate too much from a "standard" React Native setup.

We'll walk through setting up a project in the next section, Quick Start.

React Native CLI

If you're integrating React Native into an existing native app, or if you know you'll need custom native modules in your app, you'll want to use the react-native CLI to create your app. You can read how to do this on the Facebook docs for environment setup. Click on the "React Native CLI Quickstart" tab.

In terms of the JavaScript code, the main difference from expo-cli is that react-native apps require registering the root component of your app using AppRegistry.registerComponent, e.g:

This allows you to optionally have multiple React Native root components, which you might embed into different native screens.

In this class, we will use **expo-cli** to develop native applications.

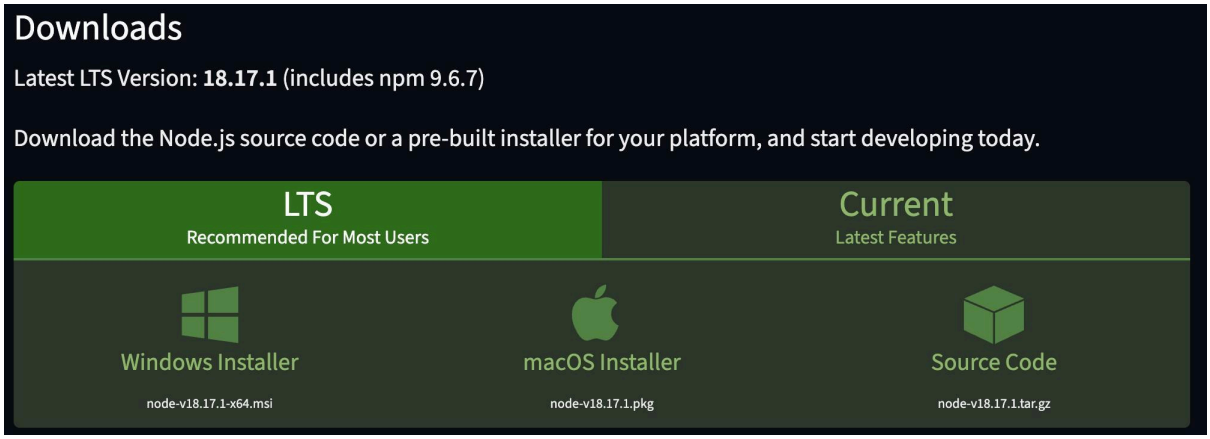
4. Materials and Equipment

To properly perform this activity, the student must have:

- Node.js LTS
- Expo
- VS Code (or any code editor).

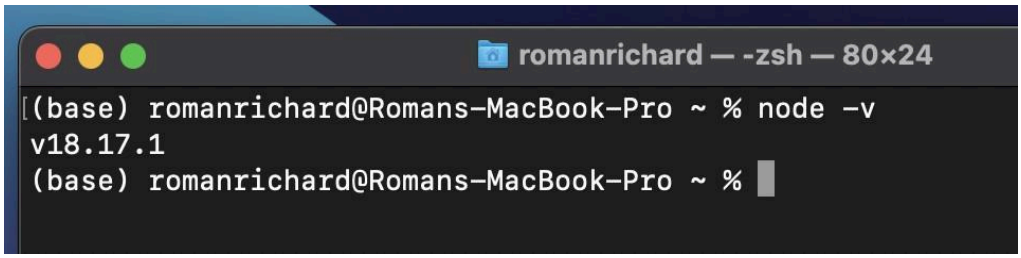
5. Procedure

1. Download node.js LTS.



It is recommended that you install the LTS version. This is intended for most users due to stability. To confirm installation, you can open your terminal and check the version of your node.js.

It should show version 18.17.1 like the figure below.



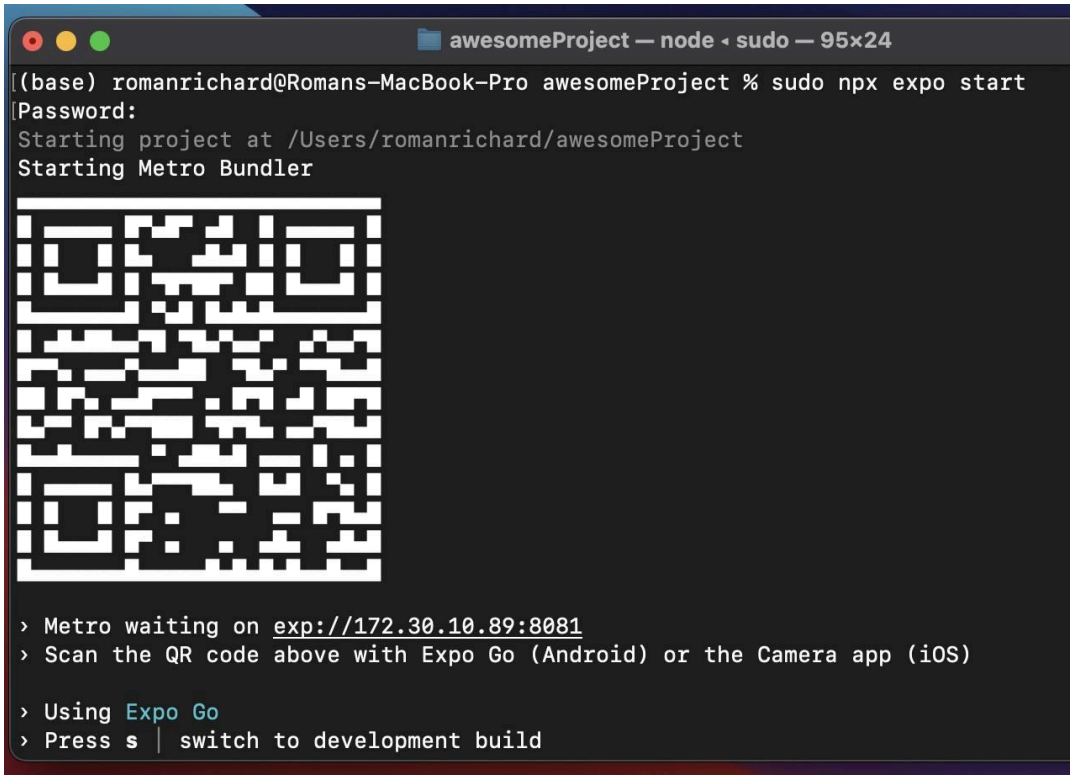
2. Create a new React Native project called "awesomeProject" using expo. Run the command *npx create-expo-app awesomeProject*

```
romanrichard — -zsh — 95×24
(base) romanrichard@Romans-MacBook-Pro ~ % sudo npx create-expo-app awesomeProject
Password:
✔ Downloaded and extracted project files.
> npm install
npm WARN deprecated @npmcli/move-file@1.1.2: This functionality has been moved to @npmcli/fs
npm WARN deprecated uuid@3.4.0: Please upgrade  to version 7 or higher.  Older versions may use
 Math.random() in certain circumstances, which is known to be problematic.  See https://v8.dev/
blog/math-random for details.
npm WARN deprecated uglify-es@3.3.9: support for ECMAScript is superseded by `uglify-js` as of
v3.13.0

added 1222 packages, and audited 1223 packages in 2m

69 packages are looking for funding
  run `npm fund` for details

5 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
```

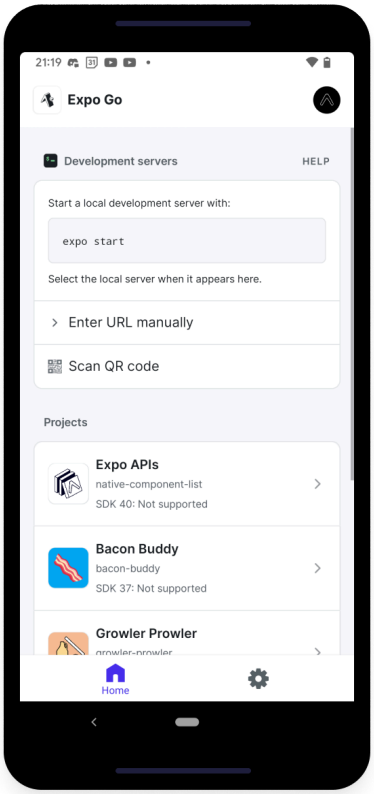
Let expo install dependencies such as shown in the figure. If you're able to achieve something similar to the screenshot below, then you can proceed to the next step.

```
✅ Your project is ready!

To run your project, navigate to the directory and run one of the following npm commands.

- cd awesomeProject
- npm run android
- npm run ios
- npm run web
(base) romanrichard@Romans-MacBook-Pro ~ %
```

3. Run expo using *npx expo start*.



Make sure to download Expo Go on your phone to scan the code shown to connect. Alternatively, you can also input the shown socket.



The following sections will have no screenshots provided as a guide, you are expected to include all your answers here under section 6: output.

4. Open VSCode (or any compatible editor) and explore the folder.
5. Provide a quick summary of the contents of the awesomeProject application.

6. Output

Provide an output of your work here.

Setup

```
C:\Users\eesj\Documents>npx create-expo-app SanJuan_SD --template blank
Creating an Expo project using the blank template.

√ Downloaded and extracted project files.
> npm install
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache i
f you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerfu
l.
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabi
lities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorb
itant rates) by contacting i@izs.me
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabi
lities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorb
itant rates) by contacting i@izs.me
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabi
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lities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorb
itant rates) by contacting i@izs.me
\
```
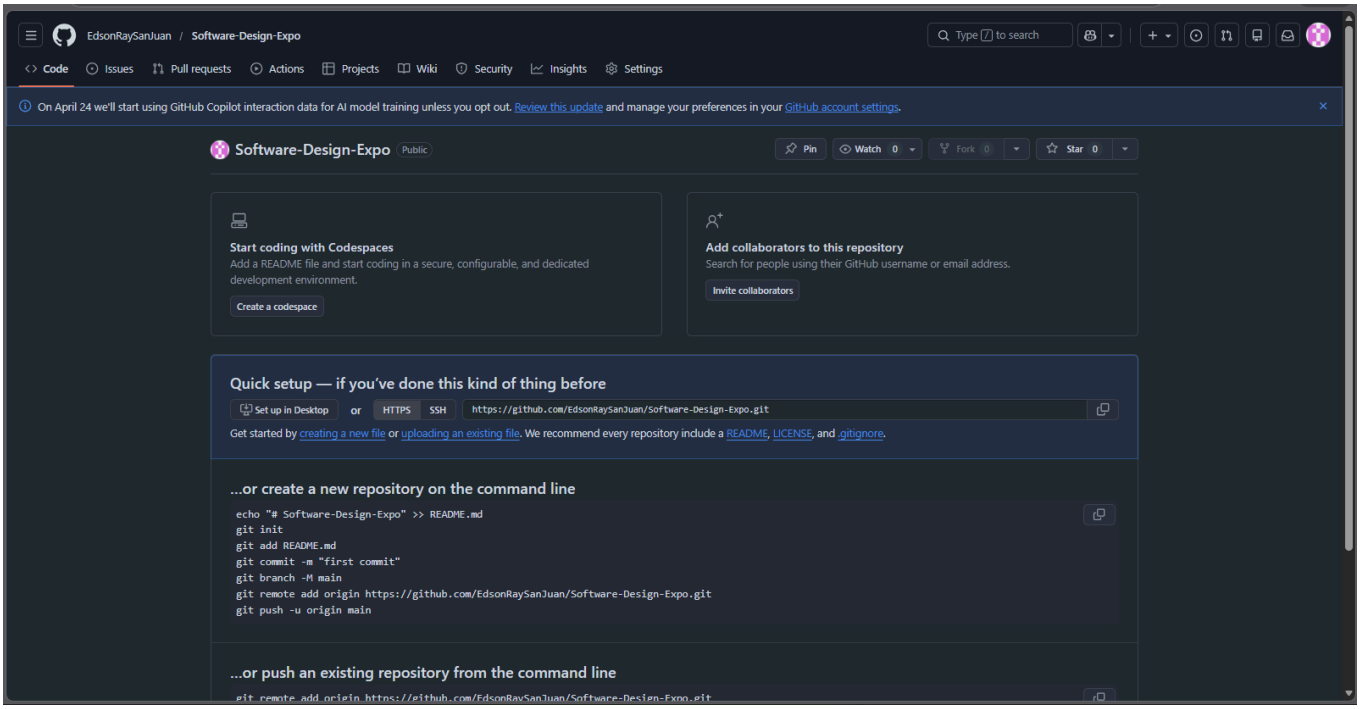

Change directory to your created folder and run vscode

```
C:\Users\eesj\Documents>cd SanJuan_SD

C:\Users\eesj\Documents\SanJuan_SD>code .

C:\Users\eesj\Documents\SanJuan_SD>
```

Creating Github Repository



Linking VSCODE to Github

```
C:\Users\eesj\Documents\SanJuan_SD>git config --global user.name EdsonRaySanJuan

C:\Users\eesj\Documents\SanJuan_SD>git config --global user.email qeresanjuan@tip.edu.ph
```

Initialize and connect the github repository in vs code

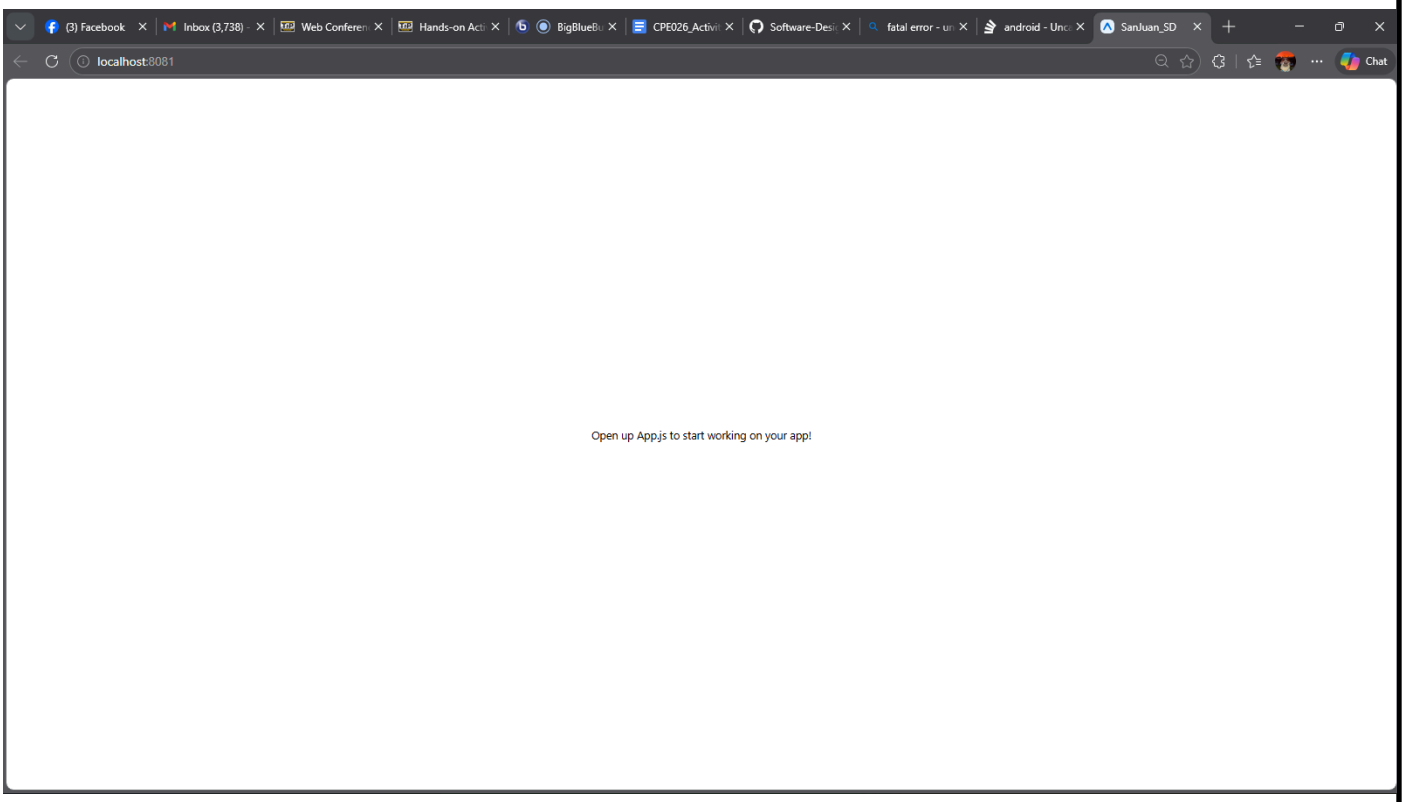
```
PS C:\Users\eesj\Documents\SanJuan_SD> echo "# Software-Design-Expo" >> README.md
>> git init
>> git add README.md
>> git commit -m "first commit"
>> git branch -M main
>> git remote add origin https://github.com/EdsonRaySanJuan/Software-Design-Expo.git
>> git push -u origin main
Reinitialized existing Git repository in C:/Users/eesj/Documents/SanJuan_SD/.git/
[master f96dfb2] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 README.md
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 12 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (18/18), 558.06 KiB | 20.67 MiB/s, done.
Total 18 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/EdsonRaySanJuan/Software-Design-Expo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\eesj\Documents\SanJuan_SD>
```

To run the App.js enter the command npx expo start

```
PS C:\Users\eesj\Documents\SanJuan_SD> npx expo start
Starting project at C:\Users\eesj\Documents\SanJuan_SD
Starting Metro Bundler

› Scan the QR code above to open in Expo Go.
› Metro: exp://192.168.254.101:8081

› Using Expo Go (Press s to switch to development build)
› Press ? | show all commands

Logs for your project will appear below. Press Ctrl+C to exit.
```

To run in the browser (Optional)

```
PS C:\Users\eesj\Documents\SanJuan_SD> npx expo install react-dom react-native-web
› Installing 2 SDK 55.0.0 compatible native modules using npm
› npm install
```



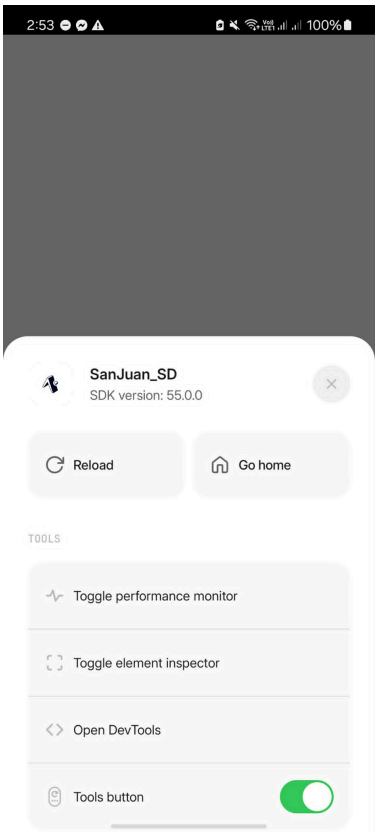
fatal error - uncaught error: java.ioexception: failed to download remote update

```
PS C:\Users\eesj\Documents\SanJuan_SD> npm install --save-dev @expo/ngrok
npm warn deprecated uuid@3.4.0: Please upgrade  to version 7 or higher.  Ol
ls.

added 32 packages, and audited 602 packages in 10s

53 packages are looking for funding
  run `npm fund` for details

18 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force
```

```
PS C:\Users\eesj\Documents\SanJuan_SD> npx expo start --tunnel
>>
>>
Starting project at C:\Users\eesj\Documents\SanJuan_SD
Starting Metro Bundler

Tunnel connected.
Tunnel ready.

› Scan the QR code above to open in Expo Go.
› Metro: exp://9n_vxfe-anonymous-8081.exp.direct
› Web: http://localhost:8081

› Using Expo Go (Press s to switch to development build)
› Press ? | show all commands

Logs for your project will appear below. Press Ctrl+C to exit.
Android Bundled 7297ms index.js (699 modules)
```

Verification in Phone that it works

Expo Go in Phone (Connected)

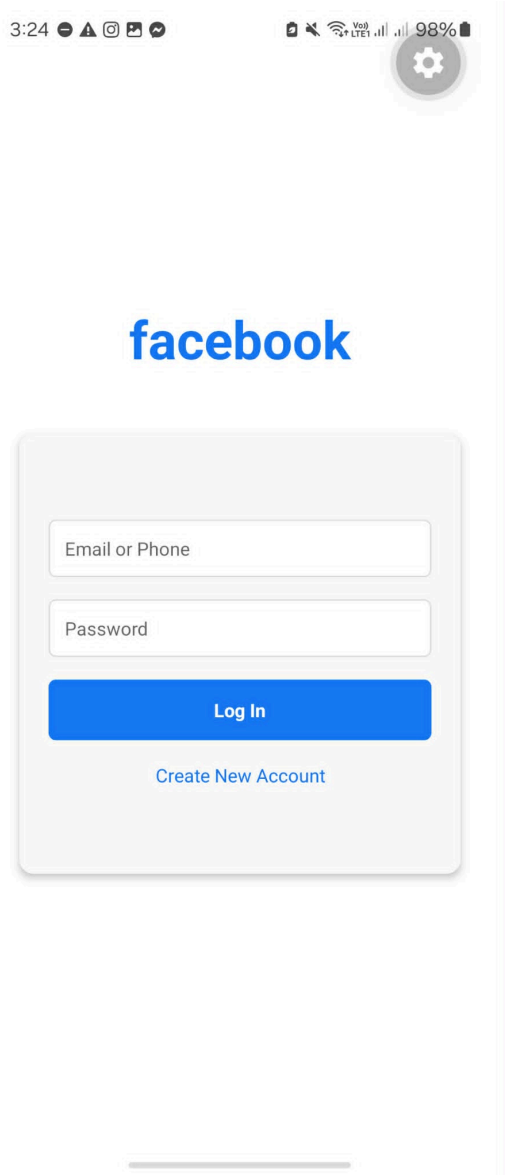


7. Supplementary Activity

Modify **app.js** to reflect all names of the members of the team. This change must then reflect on your live demo. Provide the following:

- What did you change in app.js?
  - On my modified code, I did a simple Facebook login sign-up look-alike. I placed the contents — like the rectangle card and the text inputs for username and password — inside a formContainer so they sit neatly in a square card. Then I adjusted the container's fixed dimensions, centered it, and added basic styling such as background color, padding, and rounded corners to make it resemble the Facebook login/sign-up user interface. The "facebook" title sits above the card, while the inputs, action button, and toggle link are grouped inside.

- Provide a picture of the live demo on Android and iOS device.



8. Conclusion

In conclusion, I did a simple Facebook login/sign-up look-alike. I grouped the inputs and buttons inside a formContainer to give it a fixed square dimension, then centered it, and added basic styling like background color, padding, and rounded corners. The “facebook” title sits above the card, while the username, password, confirm password, action button, and toggle link are neatly organized inside. Overall, I transformed the default layout into a clean, square card design that mimics the Facebook login page.

9. Assessment Rubric